

Structural proof of SL_γ : closing the column-1 SC discriminator for Ω on $\lambda = (3, 3, 1, 1, 1)$

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Abstract

We close the last structural gap in the column-1 SC discriminator for the transfer operator Ω on $\lambda = (3, 3, 1, 1, 1)$. The residual lemma SL_γ — that $\pi_+(R'_9 v_T) \in V_+^\alpha \oplus V_+^\beta$ for every γ -SYT T — is proved by a single positional observation: the lower chain $S_6 S_5 S_4 S_3$ permutes only the values $\{3, 4, 5, 6, 7\}$ and therefore *cannot move the entries 8 and 9*; once 8 sits at $(5, 1)$ and 9 at $(2, 3)$, the column-1 triple is forced to come from $\{3, 4, 5, 6, 7\}$ and *automatically* realises one of the patterns α, β, γ . There is no room for a survivor. The factor S_7 then annihilates the γ -component and S_8 leaves $V_+^\alpha \oplus V_+^\beta$ invariant. Combined with the previously established L_α, L_β , this completes the forward direction of the discriminator on $\lambda = (3, 3, 1, 1, 1)$.

1 Setup and recollection

We work in the Hecke algebra $H_9(q)$ with Hoefsmit's semi-normal basis $\{v_T : T \in \text{SYT}(\lambda)\}$, $\lambda = (3, 3, 1, 1, 1)$, $N = 120$. Write $S_i := T_i + 1$ ($i = 1, \dots, 8$) and

$$R'_k := S_{k-1} S_{k-2} \cdots S_2 S_1, \quad \Omega := R'_3 R'_4 R'_5 R'_6 R'_7 R'_8 R'_9.$$

For $T \in \text{SYT}(\lambda)$ let $T(r, c)$ denote the entry in row r , column c (both 1-indexed). On v_T , the generator T_i has eigenvalue q if $i, i+1$ lie in the same row of T (so $S_i v_T = (q+1)v_T$), eigenvalue -1 if they lie in the same column (so $S_i v_T = 0$), and otherwise $S_i v_T = a_i v_T + b_i v_{s_i T}$, where $s_i T$ is the SYT obtained from T by swapping the values $i, i+1$ (this swap fixes the cells of all other values).

Recall the T_1 -eigenspace decomposition $V = V_+ \oplus V_-$ with

$$V_+ := \text{span}\{v_T : T \text{ SR-at-1}\}, \quad V_- := \text{span}\{v_T : T \text{ SC-at-1}\} = \ker S_1,$$

where SR-at-1 means $T(1, 1) = 1$, $T(1, 2) = 2$ and SC-at-1 means $T(1, 1) = 1$, $T(2, 1) = 2$.

The three column-1 SC patterns (labels). For an SR-at-1 SYT T we say

$$\alpha : T(2, 1) = 3 \wedge T(3, 1) = 4; \quad \beta : T(3, 1) = 5 \wedge T(4, 1) = 6; \quad \gamma : T(4, 1) = 7 \wedge T(5, 1) = 8.$$

These conditions are not mutually exclusive (a SYT may be both α and γ), but as *sets* the spans

$$V_+^\alpha := \text{span}\{v_T : T \text{ } \alpha\}, \quad V_+^\beta := \text{span}\{v_T : T \text{ } \beta\}, \quad V_+^\gamma := \text{span}\{v_T : T \text{ } \gamma\}$$

are well defined subspaces of V_+ .

What is already proved. The companion note 2026-05-19-coll-sc-discriminator-structural.tex establishes L_α and L_β :

L_α : if T is α , then $R'_k v_T \in V_-$ for every $k \geq 4$ (in particular $R'_k(V_+^\alpha) \subseteq V_-$);

L_β : if T is β , then $\Omega v_T = 0$; moreover $\pi_+(R'_8 v_T) \in V_+^\alpha$ for β -SYTs (the R'_8 -form $\text{SL}_\beta\text{-R8}$).

The whole forward direction reduces to the single open statement below.

Theorem 1 (SL_γ). *For every γ -SYT T of $\lambda = (3, 3, 1, 1, 1)$,*

$$\pi_+(R'_9 v_T) \in V_+^\alpha \oplus V_+^\beta,$$

where $\pi_+ : V \rightarrow V_+$ is the projection along V_- .

Once Theorem 1 is in hand, L_γ (hence the discriminator) follows verbatim from the argument already recorded in 2026-05-19-... .tex (reproduced as Corollary 2 below).

2 The position of 8 and 9 is frozen

Lemma 1 (Frozen corners). *Let T be a γ -SYT. Then $T(4, 1) = 7$, $T(5, 1) = 8$ and $T(2, 3) = 9$. Moreover the operators S_3, S_4, S_5, S_6 fix the cells occupied by 8 and 9: for any SYT T' appearing with nonzero coefficient in $S_6 S_5 S_4 S_3 v_T$, one still has $T'(5, 1) = 8$ and $T'(2, 3) = 9$.*

Proof. By definition of γ , 7 is at $(4, 1)$ and 8 at $(5, 1)$. The largest entry 9 occupies an outer corner of λ ; the corners are $(1, 3), (2, 3), (5, 1)$. Cell $(5, 1)$ is taken by 8. If 9 were at $(1, 3)$ then column 3 would read $T(1, 3) = 9 > T(2, 3)$, violating column-strict increase. Hence 9 is at $(2, 3)$.

Each $S_j = T_j + 1$ acts on the basis through the value-transposition s_j , which interchanges the values j and $j+1$ and fixes every other cell. For $j \in \{3, 4, 5, 6\}$ the moved values lie in $\{3, 4, 5, 6, 7\}$, so neither 8 nor 9 is ever displaced. Consequently every SYT in the support of $S_6 S_5 S_4 S_3 v_T$ has 8 at $(5, 1)$ and 9 at $(2, 3)$. $\square$

3 No survivor can hide behind frozen corners

This is the combinatorial heart of the proof.

Lemma 2 (No-survivor). *Let T' be an SR-at-1 SYT of $\lambda = (3, 3, 1, 1, 1)$ with $T'(5, 1) = 8$ and $T'(2, 3) = 9$. Then T' is α , β , or γ .*

Proof. Fix such a T' . The cells $(1, 1), (1, 2)$ hold 1, 2 (SR-at-1); the cells $(5, 1), (2, 3)$ hold 8, 9. The remaining five cells

$$(1, 3), (2, 1), (2, 2), (3, 1), (4, 1)$$

are filled by the five values $\{3, 4, 5, 6, 7\}$. In particular the three *column-1* entries below the first row,

$$a := T'(2, 1), \quad b := T'(3, 1), \quad c := T'(4, 1),$$

form a strictly increasing triple $a < b < c$ chosen from $\{3, 4, 5, 6, 7\}$ (the other two of the five values occupy $(1, 3)$ and $(2, 2)$).

We claim that any increasing triple $a < b < c$ from $\{3, 4, 5, 6, 7\}$ satisfies

$$(a = 3 \wedge b = 4) \quad \text{or} \quad (b = 5 \wedge c = 6) \quad \text{or} \quad c = 7. \quad (\star)$$

Indeed, suppose $c \neq 7$. Being the largest of three distinct elements of $\{3, 4, 5, 6, 7\}$, c is at least 5, so $c \in \{5, 6\}$.

- If $c = 5$ then necessarily $(a, b, c) = (3, 4, 5)$, giving $a = 3 \wedge b = 4$.
- If $c = 6$ then $a < b$ are drawn from $\{3, 4, 5\}$, i.e. $(a, b) \in \{(3, 4), (3, 5), (4, 5)\}$. The first gives $a = 3 \wedge b = 4$; the latter two give $b = 5 \wedge c = 6$.

Thus $c \neq 7 \Rightarrow (a = 3 \wedge b = 4) \vee (b = 5 \wedge c = 6)$, which together with the case $c = 7$ proves $(\star)$.

Translating $(\star)$ back: $a = 3 \wedge b = 4$ means $T'(2, 1) = 3, T'(3, 1) = 4$, i.e. T' is α ; $b = 5 \wedge c = 6$ means $T'(3, 1) = 5, T'(4, 1) = 6$, i.e. T' is β ; $c = 7$ means $T'(4, 1) = 7$, and since $T'(5, 1) = 8$ this is exactly γ . Hence $T' \in \alpha \cup \beta \cup \gamma$. $\square$

Remark. Lemma 2 is the precise reason the discriminator's three strata exhaust the residual SYTs once the corners are frozen: the column-1 ladder of a 5-cell first column, with top two cells 1, * and bottom cell 8, has only the three "defect" patterns α, β, γ available to it. A survivor would need a column-1 triple avoiding all of $\{3, 4 \text{ flush at top}\}, \{5, 6 \text{ flush}\}, \{7 \text{ at } (4, 1)\}$, and no such triple exists in $\{3, 4, 5, 6, 7\}$.

Corollary 1. For a γ -SYT T , $Z := S_6 S_5 S_4 S_3 v_T \in V_+^\alpha \oplus V_+^\beta \oplus V_+^\gamma$.

Proof. The operators S_3, S_4, S_5, S_6 do not move the values 1, 2, so they preserve V_+ (they map SR-at-1 SYTs to SR-at-1 SYTs); thus $Z \in V_+$. By Lemma 1 every SYT in the support of Z has 8 at $(5, 1)$ and 9 at $(2, 3)$, and is SR-at-1; by Lemma 2 each such SYT is α, β , or γ . Hence $Z \in V_+^\alpha \oplus V_+^\beta \oplus V_+^\gamma$. $\square$

4 S_7 collapses γ ; S_8 stays put

Lemma 3 (Ladder preservation).

1. $S_7 V_+^\gamma = 0$.
2. $S_7 V_+^\alpha \subseteq V_+^\alpha$ and $S_7 V_+^\beta \subseteq V_+^\beta$.
3. $S_8 V_+^\alpha \subseteq V_+^\alpha$ and $S_8 V_+^\beta \subseteq V_+^\beta$.

Consequently $S_7(V_+^\alpha \oplus V_+^\beta \oplus V_+^\gamma) \subseteq V_+^\alpha \oplus V_+^\beta$ and $S_8(V_+^\alpha \oplus V_+^\beta) \subseteq V_+^\alpha \oplus V_+^\beta$.

Proof. (1) A γ -SYT has 7 at $(4, 1)$ and 8 at $(5, 1)$: the values 7, 8 are in the same column, so $S_7 v_T = 0$.

(2)–(3) For a basis vector $v_{T'}$ we have $S_j v_{T'} \in \text{span}\{v_{T'}, v_{s_j T'}\}$ (a multiple of $v_{T'}$ alone when $s_j T'$ is undefined). The transposition s_7 swaps the values 7, 8 and s_8 swaps 8, 9; neither touches any of the values 3, 4, 5, 6. Therefore s_7 and s_8 leave invariant both the α -condition (a statement about the cells of 3, 4) and the β -condition (about the cells of 5, 6): if T' is α then so is $s_7 T'$ (resp. $s_8 T'$), and likewise for β . Hence S_7 and S_8 map V_+^α into V_+^α and V_+^β into V_+^β .

The two displayed inclusions follow: writing $w = w_\alpha + w_\beta + w_\gamma$, $S_7 w = S_7 w_\alpha + S_7 w_\beta + 0 \in V_+^\alpha \oplus V_+^\beta$; and S_8 preserves $V_+^\alpha \oplus V_+^\beta$ by (3). $\square$

Remark. In (2)–(3) we do not claim s_7, s_8 preserve the γ -label (they do not: s_7 turns the column-stacked 7, 8 into a column-inversion, so $s_7 T'$ is undefined on a γ -SYT, consistent with $S_7 v_T = 0$). We only use that they preserve the α - and β -labels, which is immediate because those labels are conditions on values ≤ 6 .

5 Proof of SL_γ and of L_γ

Proof of Theorem 1. Let T be a γ -SYT. Since T is SR-at-1, $S_1 v_T = (q+1)v_T$, so

$$R'_9 v_T = (q+1) S_8 S_7 S_6 S_5 S_4 S_3 S_2 v_T.$$

Now $S_2 v_T = a_2 v_T + (\text{term in } V_-)$, where the second term is a multiple of $v_{s_2 T}$ (which is SC-at-1, as s_2 moves 2 off the first row into column 1) and is absent when 3 lies in row 1 of T ; the scalar a_2 is $q+1$ in that case and $q/(q+1)$ otherwise — in particular $a_2 \neq 0$. The operators $S_3, \dots, S_8$ fix the values 1, 2 and hence preserve the splitting $V = V_+ \oplus V_-$. Therefore

$$\pi_+(R'_9 v_T) = (q+1) \pi_+(S_8 S_7 S_6 S_5 S_4 S_3 S_2 v_T) = (q+1) a_2 S_8 S_7 S_6 S_5 S_4 S_3 v_T = (q+1) a_2 S_8 S_7 Z,$$

with $Z = S_6 S_5 S_4 S_3 v_T$. By Corollary 1, $Z \in V_+^\alpha \oplus V_+^\beta \oplus V_+^\gamma$. By Lemma 3, $S_7 Z \in V_+^\alpha \oplus V_+^\beta$ and then $S_8 S_7 Z \in V_+^\alpha \oplus V_+^\beta$. Since $(q+1)a_2 \neq 0$,

$$\pi_+(R'_9 v_T) \in V_+^\alpha \oplus V_+^\beta. \quad \square$$

Corollary 2 (L_γ). *If T is a γ -SYT then $\Omega v_T = 0$.*

Proof. Decompose $R'_9 v_T = w_+ + w_-$ with $w_- \in V_-$. By Theorem 1, $w_+ = \pi_+(R'_9 v_T) = w_+^\alpha + w_+^\beta$ with $w_+^\alpha \in V_+^\alpha$, $w_+^\beta \in V_+^\beta$. As each R'_k has rightmost factor S_1 , $R'_8 V_- = 0$, so $R'_8 R'_9 v_T = R'_8 w_+^\alpha + R'_8 w_+^\beta$. By L_α ($k=8$), $R'_8 w_+^\alpha \in V_-$. By SL_β -R8, $\pi_+(R'_8 w_+^\beta) \in V_+^\alpha$, i.e. $R'_8 w_+^\beta \in V_- \oplus V_+^\alpha$. Hence

$$R'_8 R'_9 v_T \in V_- \oplus V_+^\alpha.$$

Finally apply R'_7 : $R'_7 V_- = 0$, and by L_α ($k=7$), $R'_7 V_+^\alpha \subseteq V_-$. Therefore $R'_7 R'_8 R'_9 v_T \in V_-$, and the remaining factors $R'_3 R'_4 R'_5 R'_6$ (each with rightmost factor S_1) annihilate it: $\Omega v_T = 0$. $\square$

Theorem 2 (Discriminator, forward direction, on $(3, 3, 1, 1, 1)$). *For SR-at-1 $T \in \text{SYT}(3, 3, 1, 1, 1)$: if T is α , β , or γ then $\Omega v_T = 0$.*

Proof. L_α , L_β , and Corollary 2. $\square$

Together with the finite converse check on the 50 SR-at-1 SYTs (recorded in 2026-05-19-V33111-residual-dia this proves the full column-1 SC discriminator on $\lambda = (3, 3, 1, 1, 1)$ structurally.

6 Computational verification

All claims were verified by exact rational arithmetic at $q \in \{2, 3, 5, 7, 11\}$ in 2026-05-20-SLgamma-verify-num.py (parts (0) and (A) are q -independent combinatorics):

- (0) all 8 γ -labelled SYTs have 7@ (4, 1), 8@ (5, 1), 9@ (2, 3);
- (A) the 15 SR-at-1 SYTs with 8@ (5, 1), 9@ (2, 3) (6 α , 3 β , 6 γ -only; two of the α are also γ) are *all* $\alpha/\beta/\gamma$ — no survivor, Lemma 2;
- (B) $S_7 v_T = 0$ for every γ -SYT, and S_7 preserves V_+^α, V_+^β — Lemma 3(1)–(2);
- (C) S_8 preserves V_+^α, V_+^β — Lemma 3(3);
- (D) for each of the 6 γ -only SYTs: $Z \in V_+^\alpha \oplus V_+^\beta \oplus V_+^\gamma$, $Y = S_8 S_7 Z \in V_+^\alpha \oplus V_+^\beta$, and $\pi_+(R'_9 v_T)$ is a constant multiple of Y (identical support in $\alpha \cup \beta$).

The step-by-step support evolution (2026-05-20-gamma-chain-tracker.py) shows the γ -component dying *exactly* at the S_7 step and the survivor- and V_- -components being absent at *every* step, matching the proof.

7 Discussion

The proof is short because it isolates the right invariant: the *frozen corners*. Lower transpositions cannot disturb 8 and 9, and with those two corners pinned the first column has no combinatorial freedom to produce a survivor — the column-1 ladder $\{1, *, *, *, 8\}$ over the alphabet $\{3, 4, 5, 6, 7\}$ realises α , β , or γ and nothing else ($\star$). This is the “inevitability” flavour: the discriminator’s trichotomy is forced by the shape, not engineered. The earlier attempt foundered on trying to push S_7 ’s annihilating power down through S_5, S_6 via a commutation identity; the present argument instead lets $S_3, \dots, S_6$ act freely (they keep us inside $V_+^\alpha \oplus V_+^\beta \oplus V_+^\gamma$ for free) and only invokes S_7 at the top of the chain, where its action on γ is the trivial column-collapse $S_7 v_T = 0$.

Toward generalisation. The mechanism suggests the discriminator on a general SR-at-1 stratum: freeze the largest corner entries, observe that the column-1 ladder over the residual alphabet admits only the SRD “defect” patterns at the column-1 promotion indices, then collapse them top-down with the matching S_i . This is the route to the conjectured uniform statement (Open puzzle E.1 of the diagnostic).

Files

- Proof: this file.
- Verification: `~/projects/scratch/2026-05-20-SLgamma-verify.py`
- Mechanism trace: `~/projects/scratch/2026-05-20-gamma-chain-tracker.py`
- Prior lemmas: `~/projects/proofs/2026-05-19-col1-sc-discriminator-structural.tex`